

HomeAppShop

Unified Retail Management Information System



Team 10 - EFREI / AGH University

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Choice of Subdomains

Why these Subsystems?

We chose these four domains because they represent the **complete lifecycle** of a modern retail business:

- **Sales (SS-1):** Sales Management
- **Logistics (SS-2):** Order Fulfilment & Delivery
- **Supply Chain (SS-3):** Procurement & Stock Management
- **Finance (SS-4):** Reporting & Financial Management



DIFFERENT ACTORS

CLIENT

MANAGER

EMPLOYEE

BOOK KEEPER

DELIVERY SERVICE

PRODUCER

SS-1: Sales Management USE CASE SCENARIO

UC1-1 Process In-Store Sale	
Use Case ID	UC1-1
Actor(s)	Employee (primary), Client (secondary)
Goal	Register a successful in-store sale, decrement stock, and issue a receipt.
Precondition	Employee is logged in. Client is present with a chosen appliance or product code.
Trigger	Client presents the appliance or states the product code to the Employee.
Main Flow — Success Scenario	1. Employee enters the product name or code into the system. 2. System retrieves and displays: product name, price, and current stock level. 3. Employee verifies stock is available (includes UC1-3: Check Stock Availability). 4. Employee adds the product and quantity to the sale transaction. 5. System calculates the total price including applicable tax. 6. Client chooses a payment method (cash, card, or bank transfer). 7. Employee records the payment (includes UC1-4: Record Payment & Issue Receipt). 8. System decrements the stock by the sold quantity. 9. System records the transaction and generates a numbered receipt. 10. Employee hands the receipt to the Client.
Alternate / Extension Flows	• A1 — Out of stock at step 3: System shows stock = 0. Employee triggers UC1-6 (Notify Manager). Employee informs Client of delay. • A2 — Client requests home delivery after step 7: Employee initiates UC2-4 (Request Packing & Delivery). • A3 — Client adds more items: Return to step 1 for each additional item.
Postcondition	Sale recorded; inventory updated; receipt issued to Client.
Exception / Error Flows	Payment declined → transaction cancelled; stock NOT decremented; system logs the failed attempt.

SS-1: Sales Management USE CASE DIAGRAM

Team Effort

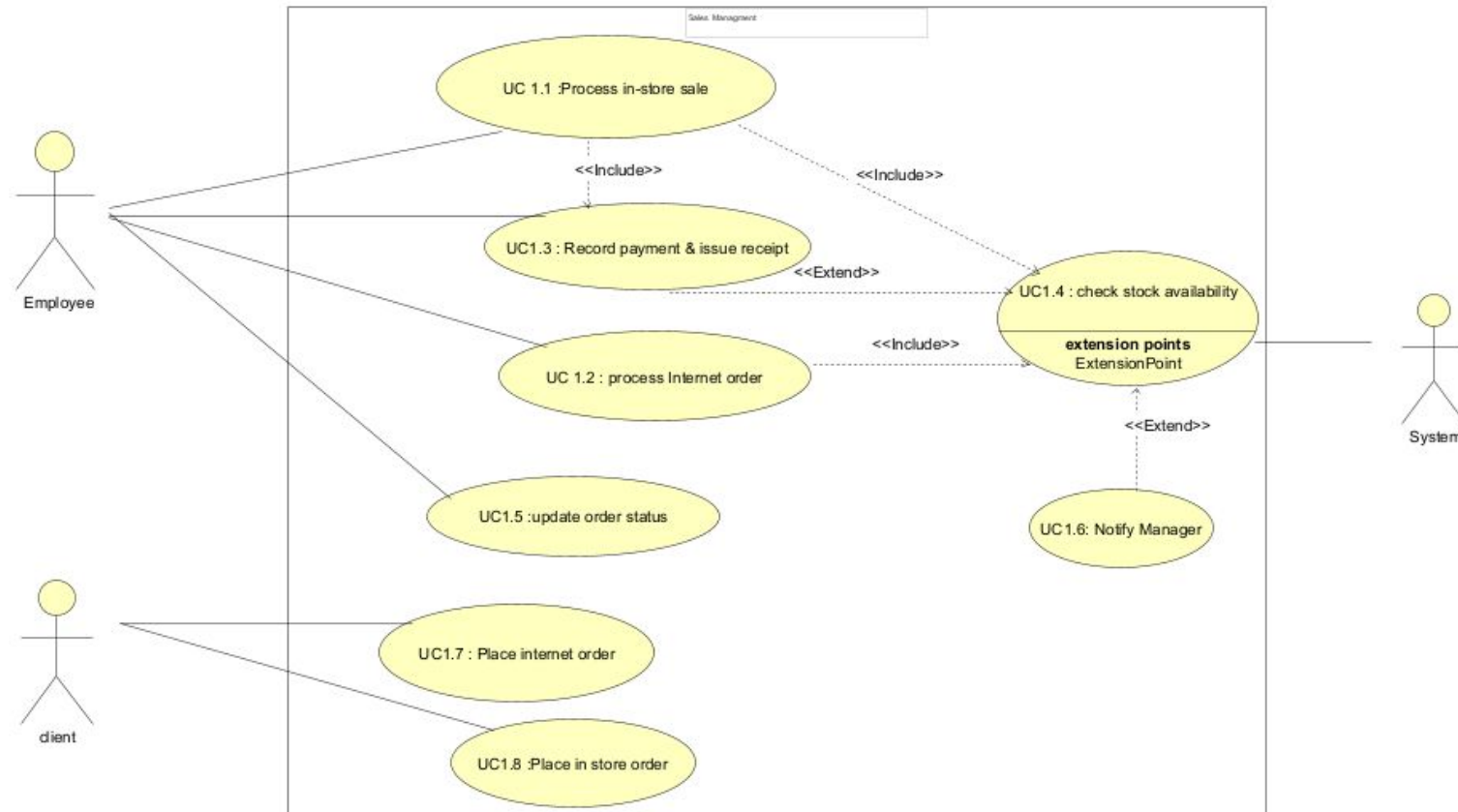


figure 1: ss1 use case diagram

SS-1: Detailed Use Cases

Team Effort

UC1-1: Process In-Store Sale

Register successful sale, update stock database, and print physical receipt for walk-in client.

UC1-2: Process Internet Order

Fetch online queue, verify payment status, and initiate fulfillment subsystem for dispatch.

UC1-3: Check Stock Level

Real-time query ensuring zero sales confirmed for unavailable appliances.

UC1-4: Record Payment

Archive transaction ID, payment method, and legal tax amount into TransactionDB.

SS-1: UML Process & Dynamics

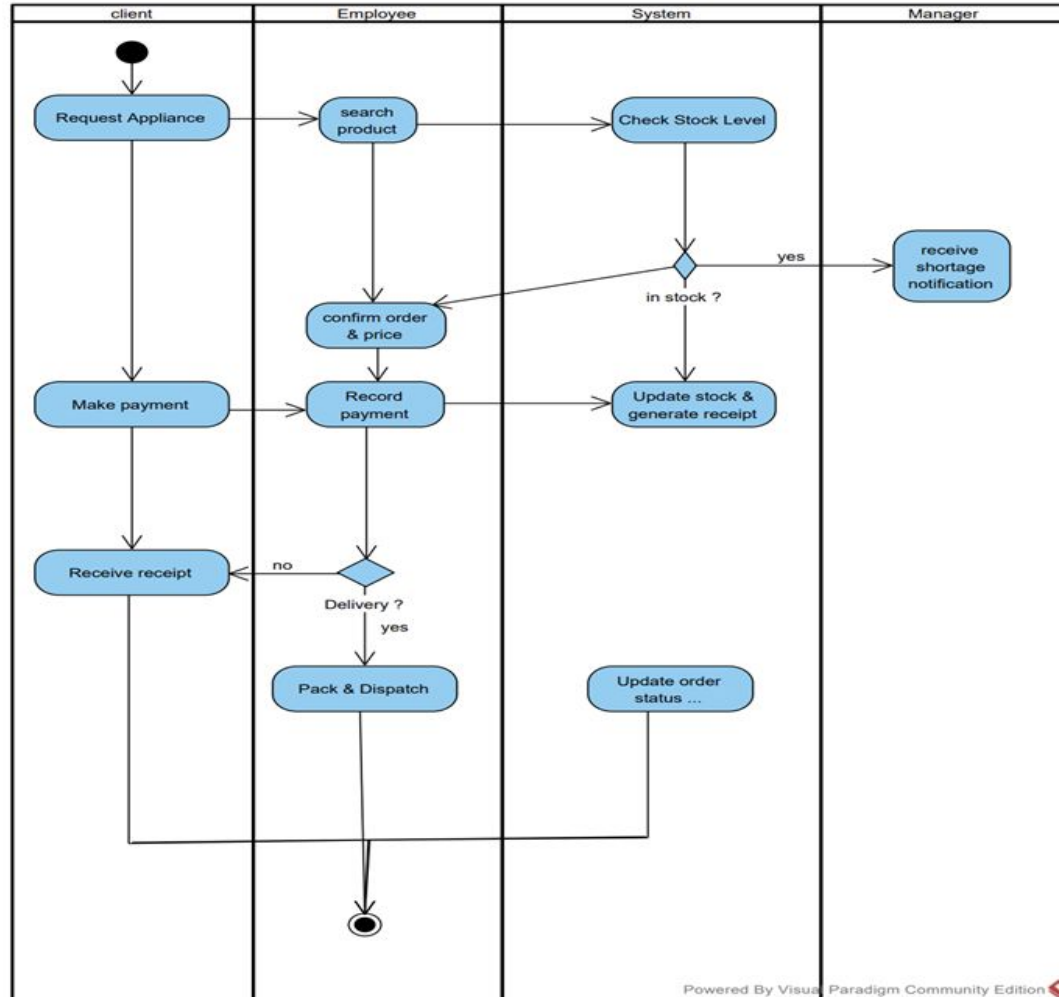


figure 2 SS1 Activity diagram

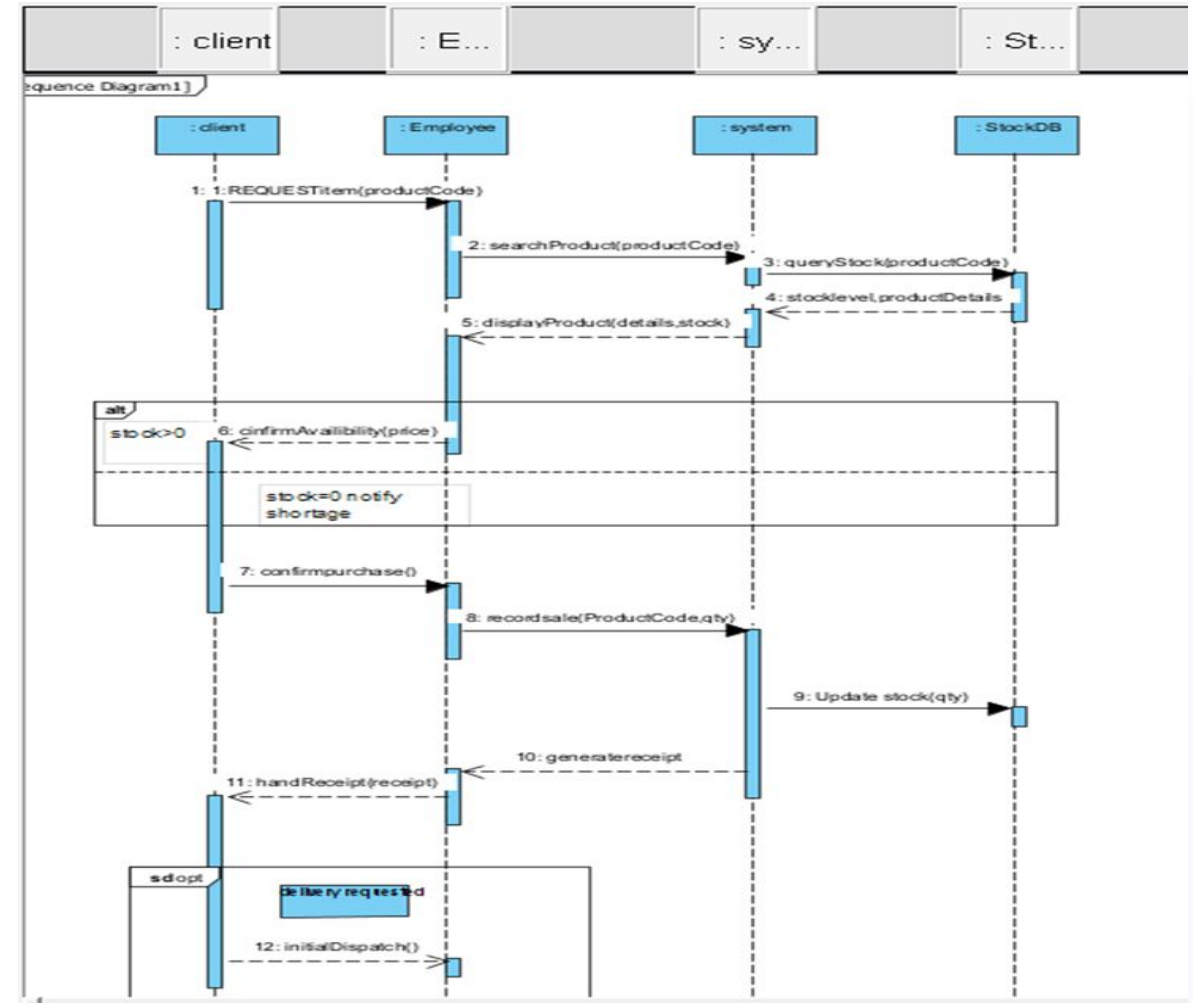


figure 3 SS1 Sequence diagram

SS-2: Order Fulfillment & Delivery USE CASE SCENARIO



UC2-1 Browse & Select Appliance	
Use Case ID	UC2-1
Actor(s)	Client (primary), System (secondary)
Goal	Allow the Client to browse the catalogue, find an appliance and view its details before ordering.
Precondition	System is online. Catalogue contains at least one active product.
Trigger	Client opens the catalogue or enters a search term.
Main Flow — Success Scenario	1. Client browses categories or enters a search keyword. 2. System retrieves and displays matching appliances with name, photo, price, and stock status. 3. Client applies optional filters (category, price range, availability). 4. System updates the list in real time. 5. Client selects an appliance to view full details (price, specs, photos, stock level). 6. If stock > 0: Client can proceed to place an order (UC2-2 or UC2-3). 7. If stock = 0: System displays 'Currently unavailable' and offers a stock alert subscription.
Alternate / Extension Flows	• A1 — No results found: System suggests related categories. • A2 — Client subscribes to stock alert: System records the subscription and notifies Client when item is restocked (UC3-5).
Postcondition	Client has viewed appliance details and can proceed to order.

SS-2: Order Fulfillment & Delivery



Lyne Ngouowa

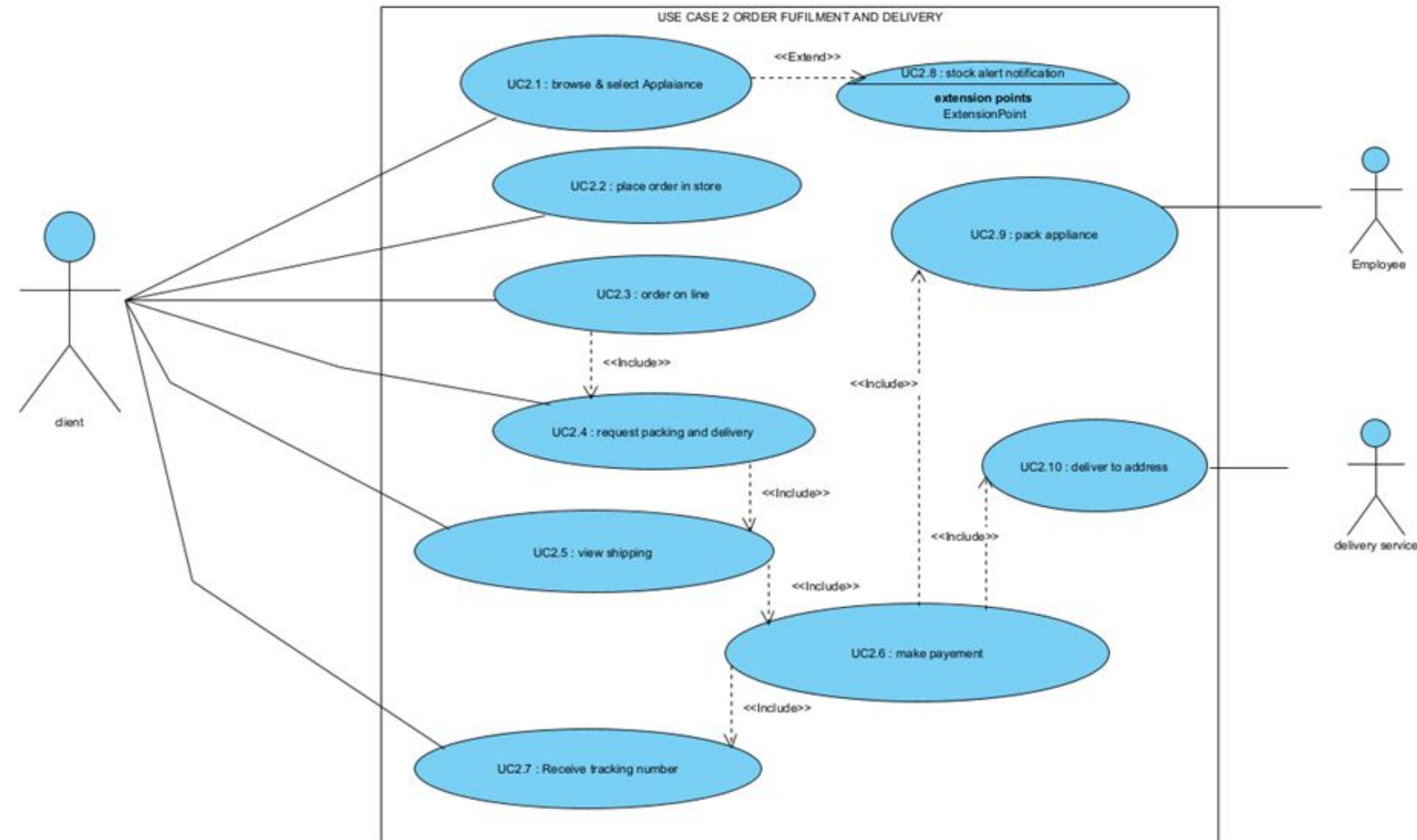


figure 4 SS2 use case diagram

SS-2: Detailed Use Cases

UC2-1: Browse & Select

Access catalog with filters (category/price) and real-time "Restock Alert" subscription.

UC2-3: Place Internet Order

The core customer journey: cart management, address entry, and secure checkout.

UC2-4: Request Delivery

System calculates fees based on weight/dimensions and assigns a Tracking Number.

UC2-5: Track Shipment

External API integration to show current status: Dispatched, In Transit, or Delivered.

SS-2: Logistics Process

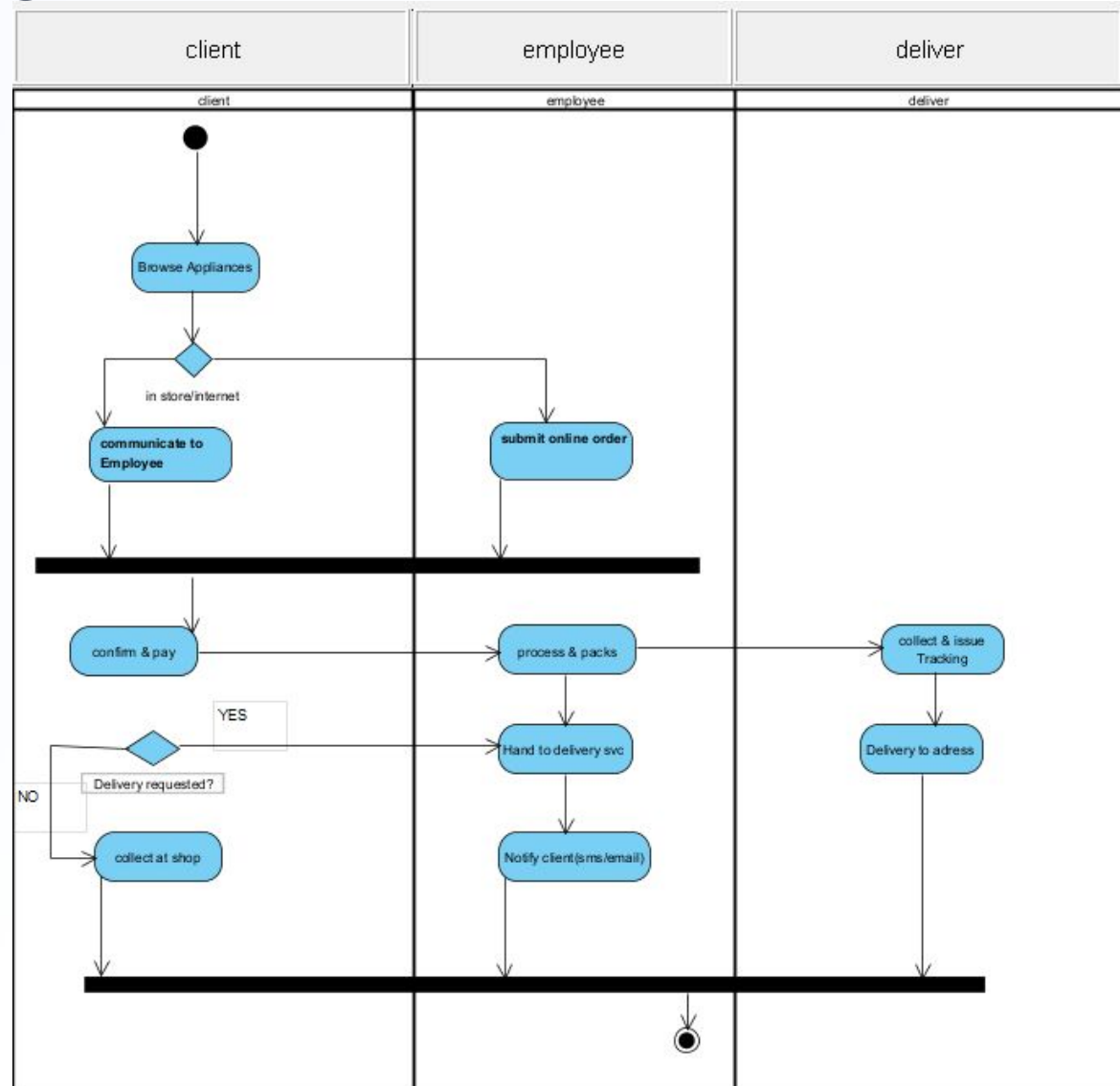


figure 5 SS2 Activity diagram

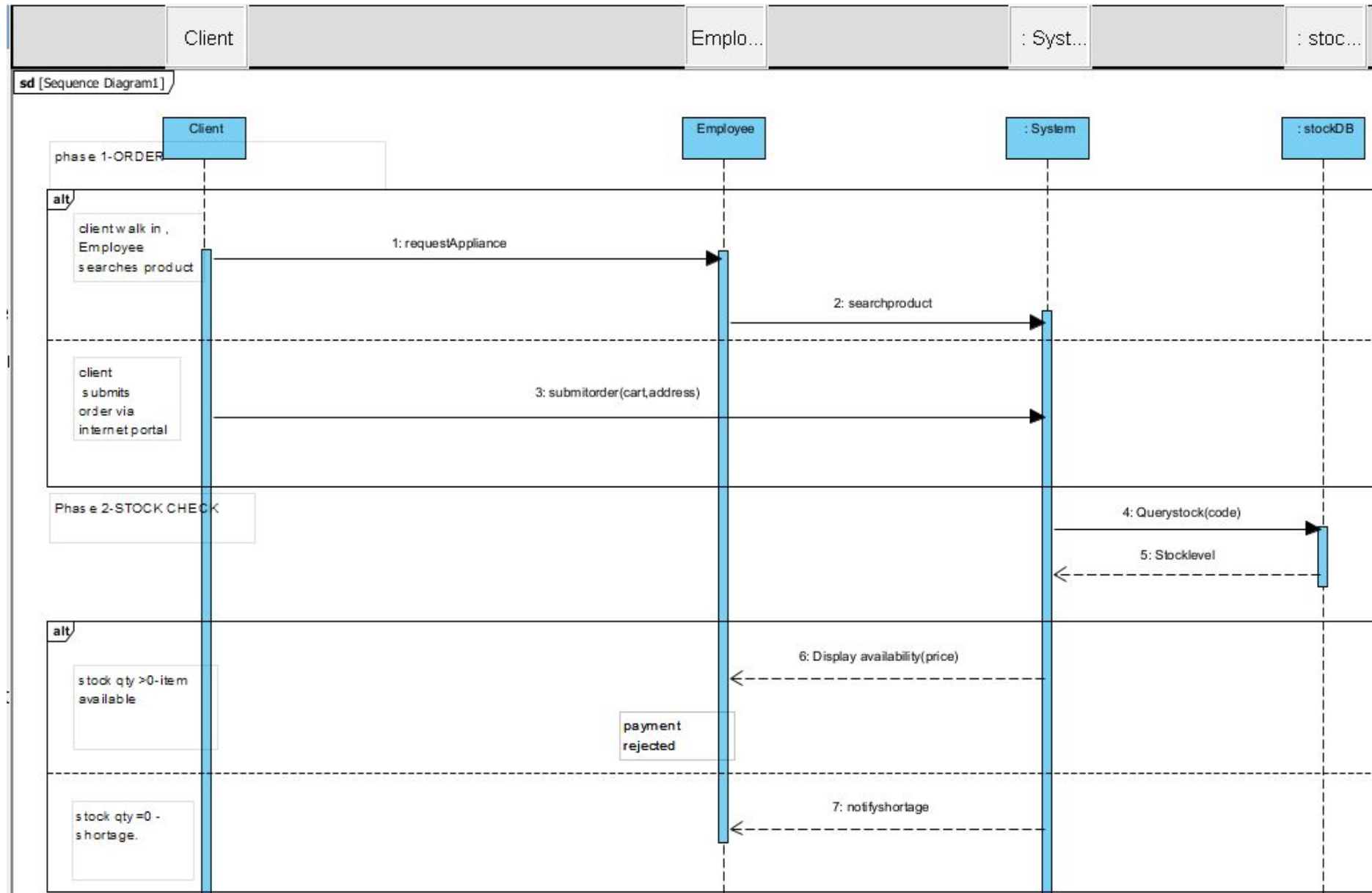


figure 6 SS2 sequence diagram

SS-3: Procurement & Stock USE CASE SCENARIO

UC3-1 Receive Stock Shortage Notification	
Use Case ID	UC3-1
Actor(s)	System (primary, automatic), Manager (recipient), Employee (alternative trigger)
Goal	Ensure the Manager is immediately and fully informed when an appliance reaches zero stock.
Precondition	Stock of an appliance has just reached 0 following a sale or manual check.
Trigger	System detects stockQty = 0 after a stock decrement. OR Employee manually reports a shortage.
Main Flow — Success Scenario	1. System detects that an appliance's stock quantity has reached 0. 2. System creates a StockAlert record: appliance name, code, quantity needed, pending Client orders. 3. System sends a notification to the Manager (dashboard alert + e-mail). 4. Appliance is automatically marked as 'Out of Stock' in the catalogue. 5. Manager receives the notification and reviews shortage details (triggers UC3-2).
Alternate / Extension Flows	• A1 — Employee manually reports shortage: Employee selects 'Report Shortage'; system creates alert and notifies Manager.
Postcondition	StockAlert recorded; Manager notified; appliance marked 'Out of Stock' in catalogue.
Exception / Error Flows	Notification delivery failure → alert remains in Manager dashboard; system retries sending.

SS-3: Procurement & Stock

Supply Chain Stability

I focused on ensuring the business never stops by automating the replenishment cycle with producers.

Key User Stories:

- **Manager:** receive auto-notification on zero stock.
- **Manager:** compare lead times of different suppliers.
- **Employee:** get notified when stock is finally available.

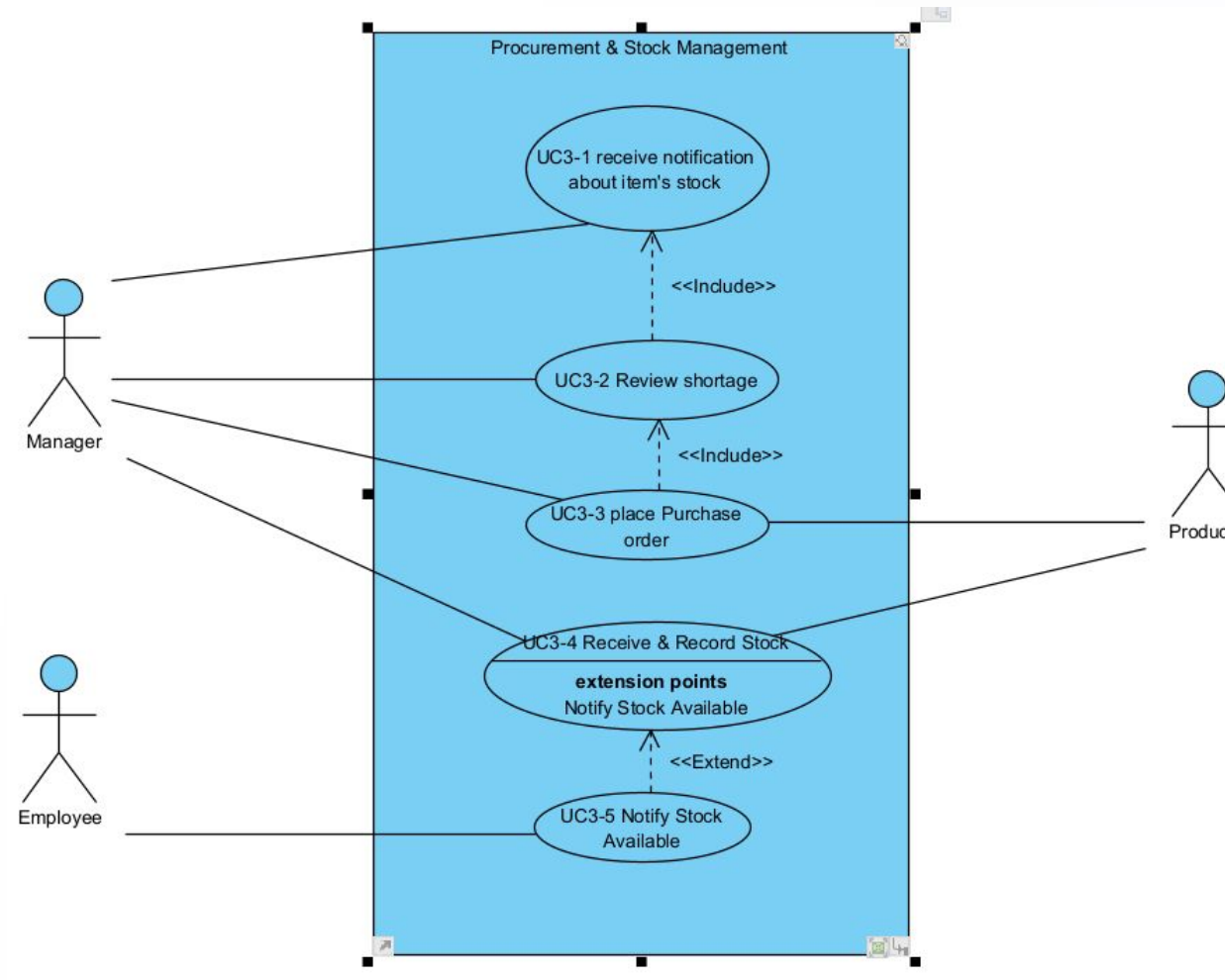


figure 7 SS3 use case diagram

SS-3: Detailed Use Cases

UC3-1: Receive Shortage Alert

Automated trigger creating a 'StockAlert' object linked to pending customer orders.

UC3-3: Place Purchase Order

Manager formalizes procurement order to the selected external warehouse or producer.

UC3-4: Confirm Receipt

Inventory update upon physical delivery of goods to the shop shelves.

UC3-5: Notify Restock

System automatically triggers emails to employees to fulfill on-hold orders.

SS-3: Restocking Process

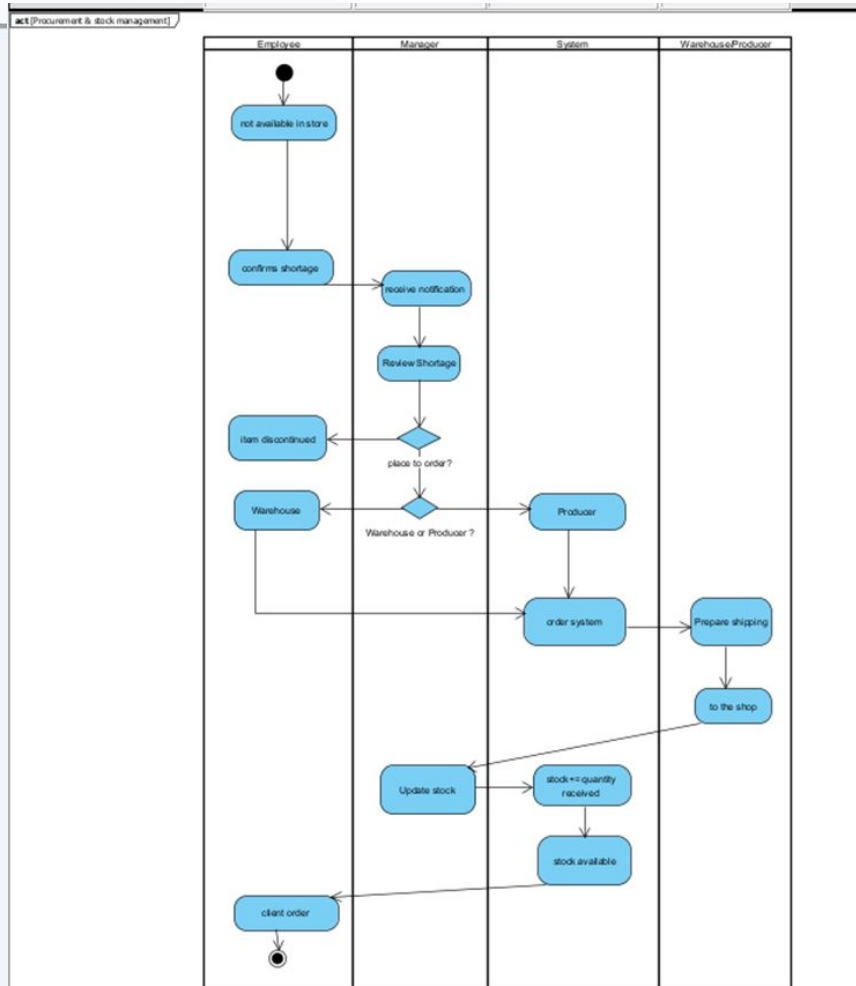


figure 8 SS3 Activity diagram

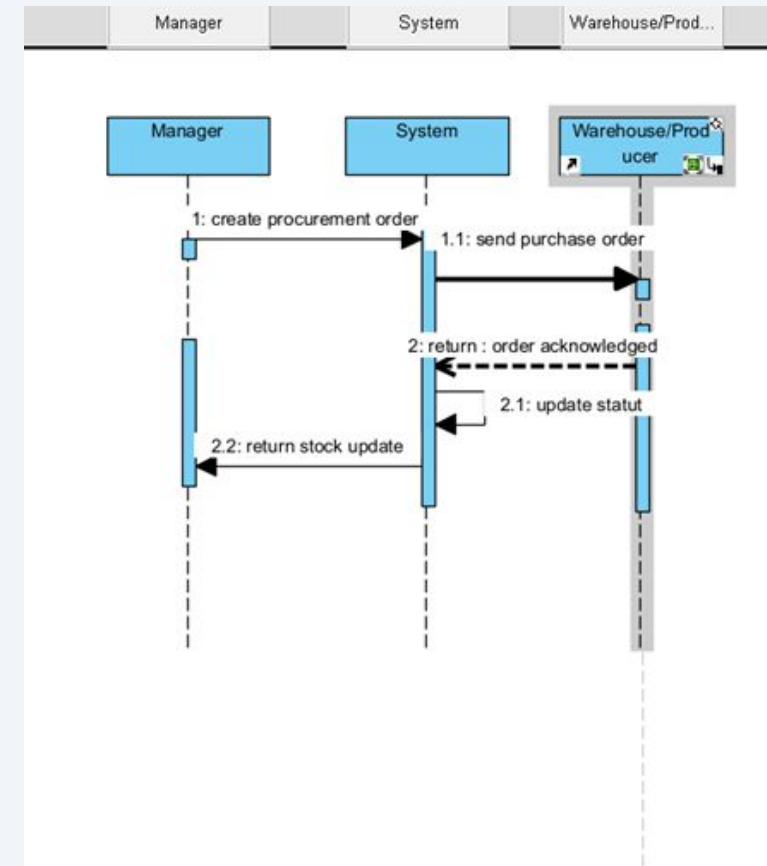


figure 9 SS3 sequence diagram

SS-4: Reporting & Financials USE CASE SCENARIO

UC4-1 Generate Monthly Sales Report	
Use Case ID	UC4-1
Actor(s)	Manager (primary), System (secondary)
Goal	Produce a complete, accurate monthly sales report for the selected calendar period.
Precondition	Manager is logged in with report-generation privileges. Sales transactions exist for the target period.
Trigger	Manager selects 'Generate Report → Monthly Sales' and specifies month and year.
Main Flow — Success Scenario	1. Manager selects the reporting period (month and year). 2. System queries all sales transactions for the period. 3. System aggregates: total revenue, number of transactions, top-selling appliances by quantity and value, revenue by category. 4. System formats data into the standard monthly sales report template. 5. System displays a preview of the report. 6. Manager reviews and confirms the report is correct. 7. Manager saves the report (includes UC4-3: Store Report in System). 8. System stores the report with date/time stamp and unique report ID.
Alternate / Extension Flows	• A1 — No transactions in period: System generates a 'Zero Sales' report; stores as per UC4-3. • A2 — Manager requests export: System offers PDF or Excel download.
Postcondition	Monthly sales report stored in system; available for transfer via UC4-4.
Exception / Error Flows	Database query timeout → Manager notified; system retries after 5 minutes.

SS-4: Reporting & Financials

Managerial Insight

We modeled the "Big Picture" – providing the financial data required to keep the business profitable and legal.

Key User Stories:

- **Manager:** generate monthly revenue reports.
- **Manager:** monitor procurement costs vs sales.
- **Manager:** transfer documents to the Book Keeper.

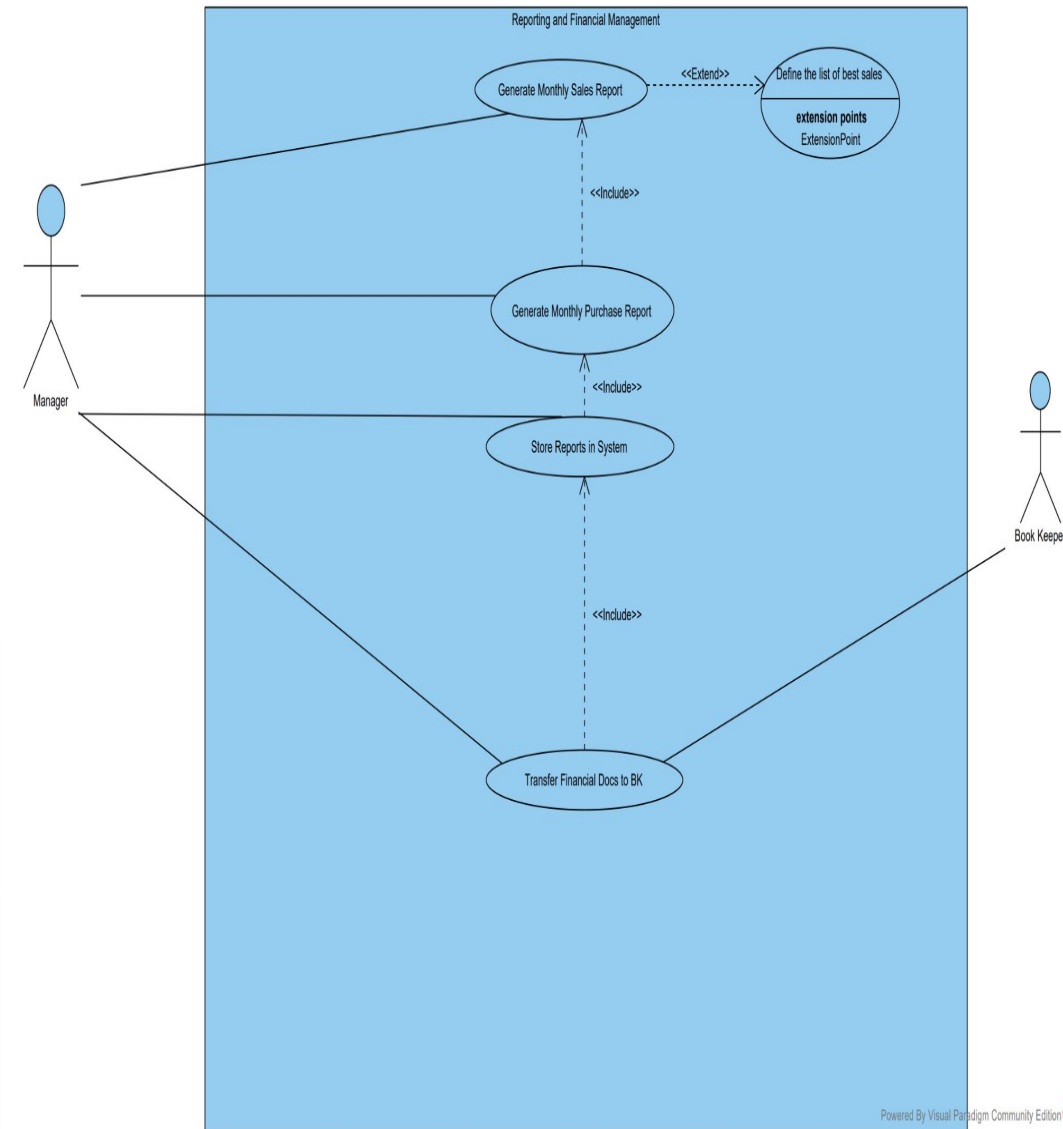


FIGURE 10 Primary actor: Manager (left). Secondary actor: Book Keeper (right, external) | [UC IDs: UC4-1 Generate Monthly Sales Report; UC4-2 Generate Monthly Purchase Report; UC4-3 Store Report in System; UC4-4 Transfer Financial Documents to Book Keeper] | [Author: Luc Lajie]

SS-4: Detailed Use Cases

UC4-1: Sales Report

Aggregate all transactions from the period to identify best-sellers and total revenue.

UC4-2: Purchase Report

Tracking the outflow of capital to suppliers for better cost management.

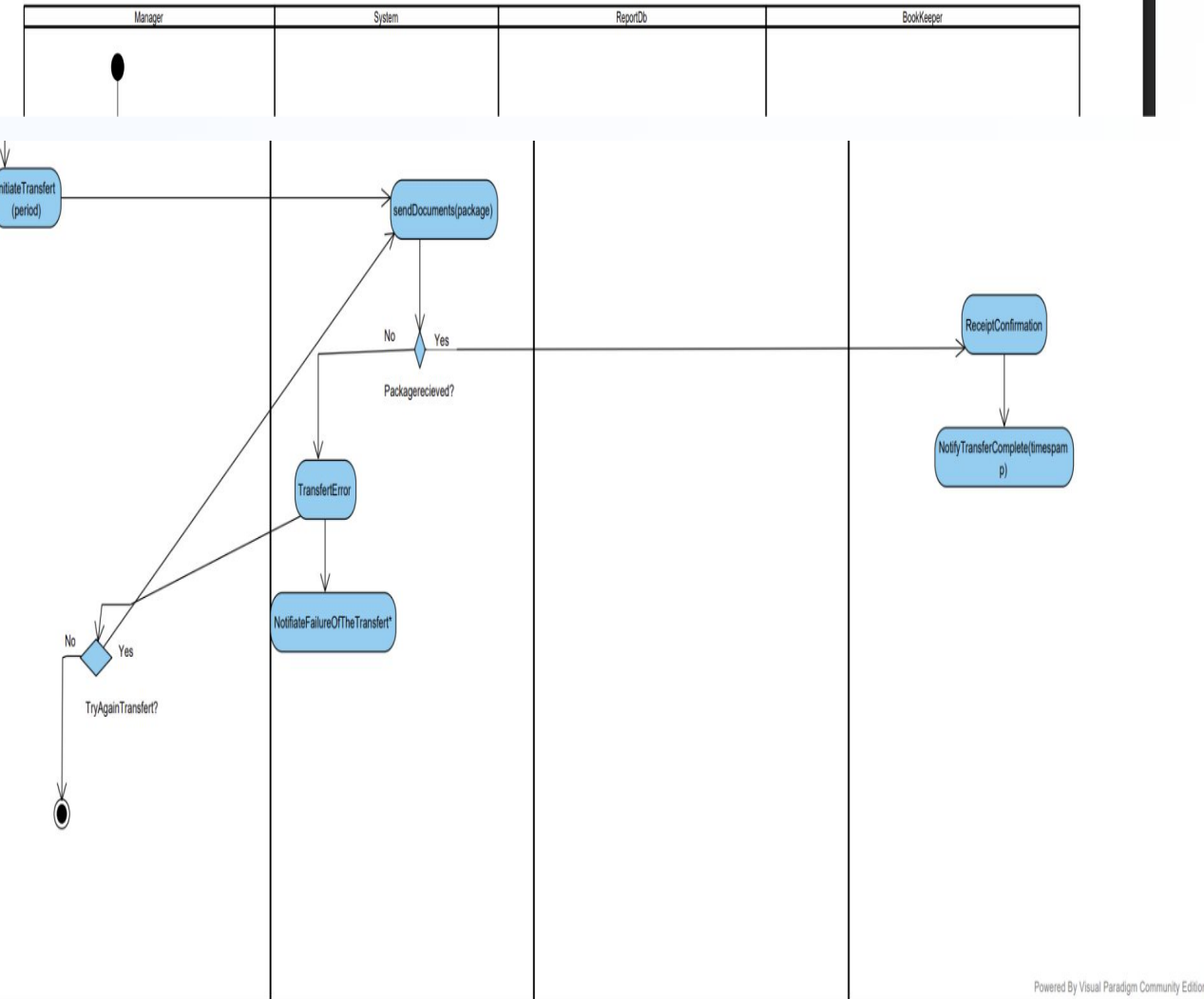
UC4-3: Document Storage

Persistence layer: Archiving PDF/Excel reports with timestamps for auditing.

UC4-4: Book Keeper Transfer

The final step: secure hand-off of all financial documents to the external accountant.

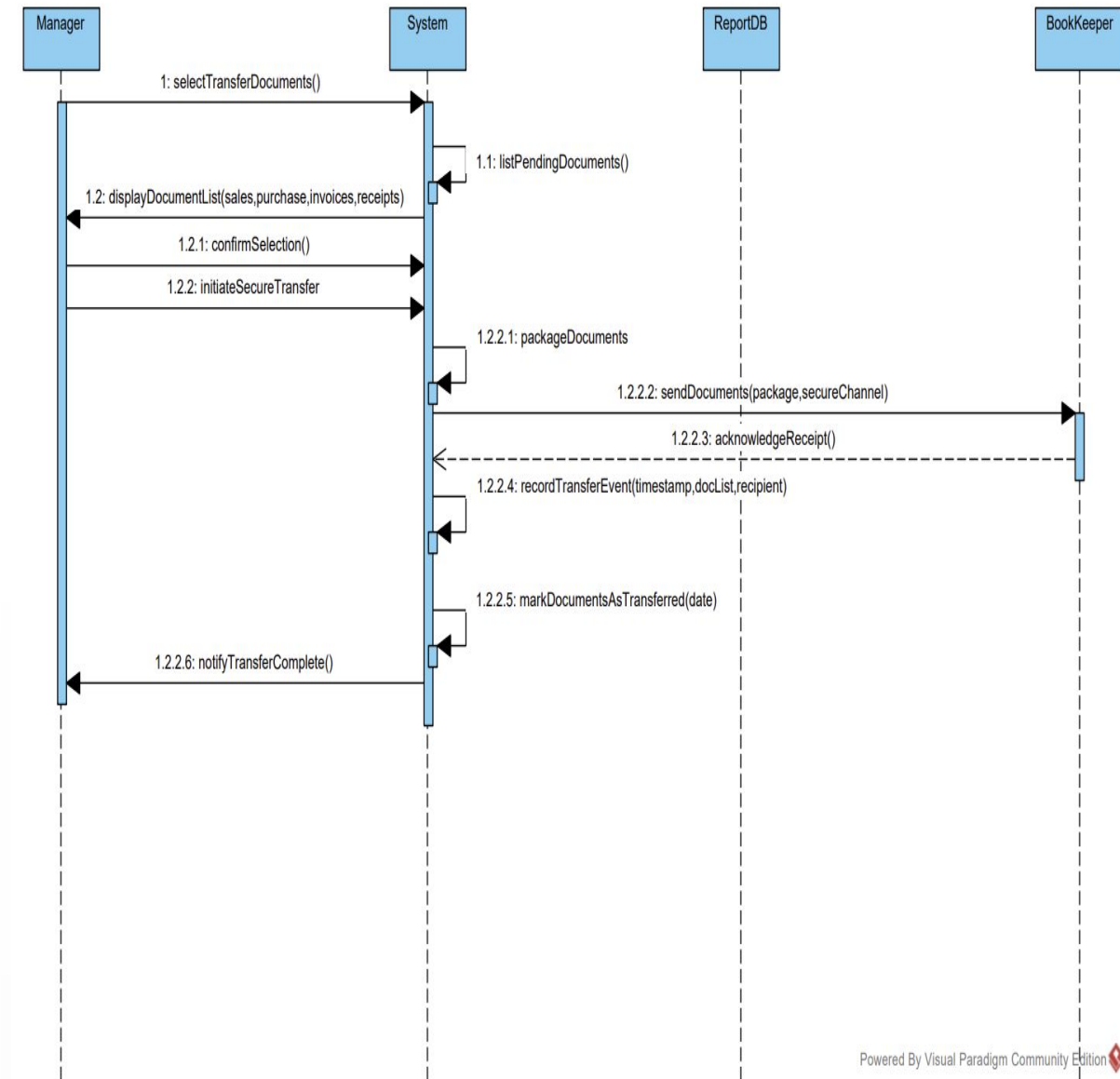
SS-4: Financial Data Flow



AD-4: Accounting Workflow

[Parallel generation of Sales and Purchase logs]

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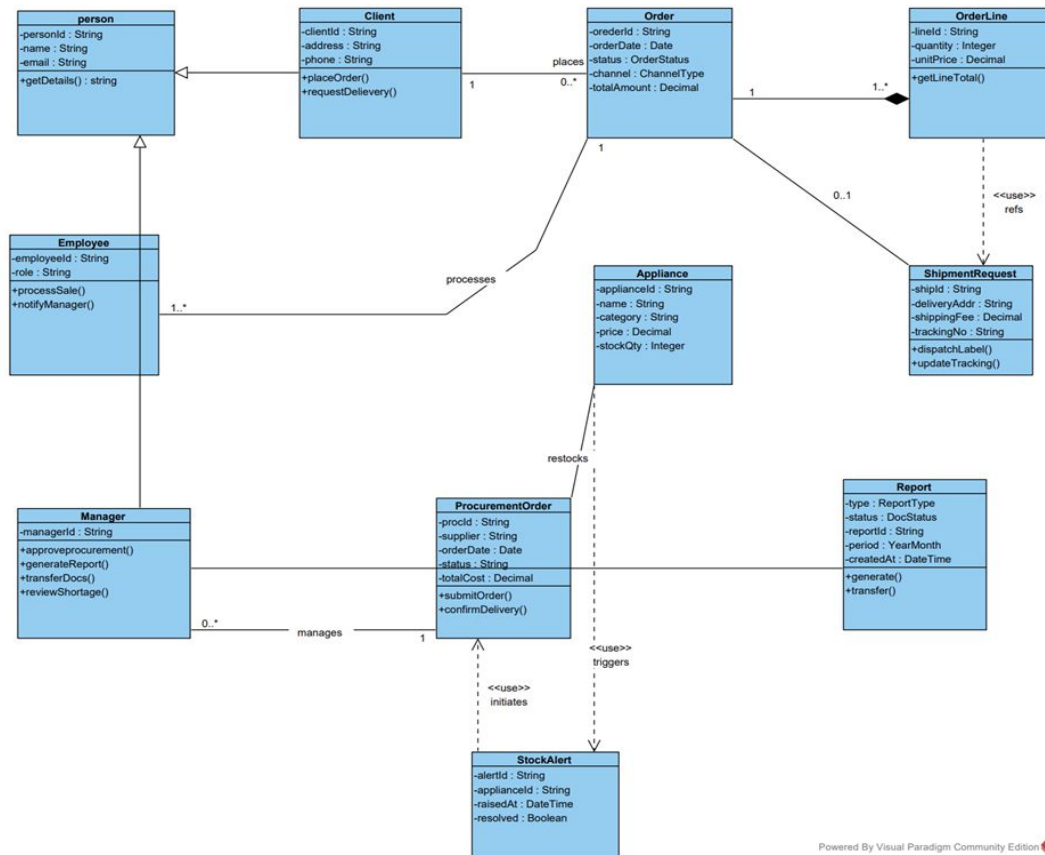


SeqD-10: UC4-1 Generate Monthly Sales Report | [Lifelines: :Manager, :System | :TransactionDB | :ReportDB] | [Author:

Luc Lajie]

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The Unified Structural Model



Core Architecture

Generalization: Person acts as abstract base for Employee, Manager, and Client.

Transactional Integrity: Order and OrderLine composition ensuring items are never orphans.

Cross-System Links: Appliance is the pivot connecting Sales, Stock, and Procurement.

FIGURE 20: CLASS DIAGRAM

Project Synthesis

Accomplishments

The HomeAppShop model demonstrates a fully integrated retail system that handles:

- End-to-end sales cycles (SS-1)
- External logistics integration (SS-2)
- Supplier chain resilience (SS-3)
- Financial transparency (SS-4)

Difficulties

During this project we came through some difficulties:

- we had to divide the system into 4 while we are 3
- Differentiate sales management and ordering fulfilment subdomain
- put all our entities in our class diagram

CONCLUSION

Thank you for your attention!